MiniProject 2

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Abstract

In this project, we studied the performance of linear classification models, namely multinomial naive Bayes and logistic regression on two textual datasets for binary and multiclass classification. Although the logistic regression model outperformed multinomial naive Bayes on the IMDb dataset in terms of accuracy by 5% (87% vs 82%), both models achieved similar accuracies when applied on the 20newsgroup dataset with about 70%. Notwithstanding, it is worth mentioning that the naive Bayes model is computationally more expensive than the logistic regression model, making logistic regression faster and more energy-efficient classifier. We used 5-fold cross-validation to tune the hyperparameters of our models on both datasets to achieve the best performance. We compared the models implemented from scratch to Scikit learn models where applicable. For instance, our logistic regression model showed better accuracy for binary classification on the IMDb dataset than the Scikit-learn model.

Furthermore, we explored the impact of using subsets of training data and features on the accuracy of our models. We observed only slight changes in the models' accuracy as a smaller set of features was utilised. However, while decreasing the number of features improved the models' performance on the IMDb dataset, it somewhat decreased the performance on the 20newsgroup dataset. We found that the performance of the models on the 20newsgroup dataset depended more heavily on the amount of information they were trained on compared to the IMDb dataset. For instance, while enlarging the size of data the models were trained on increased the accuracy on both datasets, this gain in accuracy was more significant on the 20newgroup dataset. These findings are consistent with the results in [1]. To further evaluate our models, performance metrics such as the ROC curve and confusion matrix were studied.

1 Introduction

Although information is available in many forms, text is probably the largest repository of human knowledge. Nevertheless, analyzing huge amounts of unstructured text data from emails, social media conversations, chats, and reviews is extremely challenging. Not only manually processing an enormous amount of text data is time-consuming, but it is expensive, error-prone, and tedious. Text analysis can have a range of methods and techniques including text classification, clustering, word sentence disambiguation, to name but a few. Text classification itself has a wide range of applications in sentiment analysis, intent detection and topic modelling.

In this domain, preprocessing techniques are required for converting these documents to structured documents [2]. Before converting raw textual data to numerical data, we followed some steps to clean and preprocess the datasets effectively. We worked with the bag-of-words model which is a simplifying representation in natural language processing commonly used in methods of document classification where the occurrence of words are used as features for training a classifier.

Linear classification models are used for multiclass and binary classifications. Identifying the weight of a word in a text and how much is the existence importance of that word in the text is evaluated by the number of times that word is seen in the text. TF-IDF is a statistical measure enabling us to scale the weight of a word in the text. It was demonstrated in [3] that using TF-IDF scores instead of raw word frequencies improved the performance of multinomial naive Bayes.

In this work, we utilized multinomial naive Bayes and linear regression models. We concluded that the linear regression model is more accurate in performing binary classification on the IMDb reviews dataset whereas, multinomial naive Bayes outperforms logistic regression for multiclass classification on the 20newsgroup dataset. It is noteworthy that rather than using other naive Bayes methods such as Gaussian naive Bayes and Bernoulli naive Bayes, we used multinomial naive Bayes, a logarithm based classifier, which is more suitable for the classification of textual data [1, 4]. The results of implemented models on the IMDb reviews dataset are comparable with outcomes in [5].

2 Datasets

Two datasets are subjects of analysis in this project: "Twenty Newsgroups text dataset" and "Large movie IMDB review dataset". Twenty Newsgroups is a standard dataset fetched from the sklearn library, containing 18846 samples (posts) and 20 classes (different newsgroups). This dataset is divided into 11314 training and 7532 testing samples. The IMDb review dataset encompasses 50k samples in which the training set and testing set are equally split. The training set is divided into positive and negative reviews, in other words, binary sentiment polarity labels where a negative review has a rating smaller

or equal to 4 out of 10, a positive review has a rating greater or equal to 7 out of 10 and reviews with more neutral ratings are not included.

To better understand the datasets, we visualized class distributions, most common n-grams, word clouds, etc. For example, positive and negative IMDb reviews word clouds are illustrated in Figure 1 (a) and (b). Additional dataset visualizations can be found in the submitted Notebook.

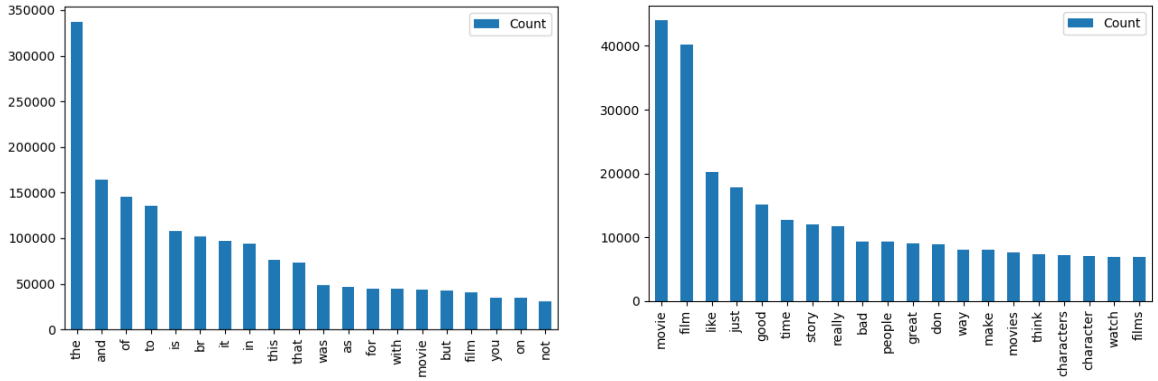


(a) Postive reviews (b) Negative reviews

Figure 1: IMDb reviews sentiment wordclouds

To begin the preprocessing stage, the 20newsgroup dataset is loaded from Scikit-learn while the IMDb dataset was constructed from raw text files. The constructed IMDb reviews dataset was then saved to be loaded on the next program runs due to this process being time-consuming. In this step, we assure that both datasets are in a similar format, enabling us to feed them into the same preprocessing subroutine. The 20newsdataset is cleaned from headers, footers and quotes. Furthermore, we replace all punctuations and delimiters such as tabs, endlines, and carriage returns with space delimiters.

After this initial cleaning step, we analyzed the most frequent words appearing in datasets, as shown in Figure 2(a). Despite the frequent use of these words in natural language, a large number of them give us no or low value information in training our models. These words referred to as stop words (such as "the", "and", "of", and to") are filtered out of our training data. Other measures taken in this step to further filter out trivial information from our training set include lower casing all characters as well as removing accents, numbers, special characters. Figure 2(b) depicts the most frequent words appearing in the IMDb reviews dataset after taking the mentioned steps.



(a) (b)

Figure 2: IMDb reviews dataset before (a) and after (b) feature engineering and removing stopwords

Taking a closer look at Figure 2(b), it can be observed that the word "character" and "characters" are considered to be distinct. However, both of these words give us the same information. This is only a good case in point. A solution to this problem would be stemming or lemmatization, as words with one stem (family word) need to be reduced to a common base form. Stemming chops off the ends of words, whereas lemmatization determines the base of the word. Among these techniques, we use lemmatization as a more reliable way to reduce inflectional and derivationally related forms of a word. To verify this claim, we compared accuracies by deploying both techniques using the natural language toolkit (NLTK) which resulted in a better outcome for lemmatization.

Once the raw text files have been feature engineered, cleaned, and lemmatized, they need to be transformed in the form of feature vectors to be appropriate for statistical analysis. The bag of words method is used to generate vectors containing the occurrences of each word in documents. After obtaining word frequencies in each document, those which occur numerous times in many samples are considered less informative. For this reason, we downscale those frequencies using the TF-IDF statistical measure. Using this method, in turn, lessens the influence of repetitive less informative words. Since each document contains only a few words, there will be many zeros in our matrix. Using a dense matrix will be extremely memory expensive,

as our matrix contains a large number of zeros, we input sparse matrices to our models for training.

3 Results

Two classification models particularly multinomial naive Bayes and logistic regression are applied to both datasets. We wrote the multinomial naive Bayes and logistic regression models from scratch, able to perform multiclass and binary classification respectively. To tune the hyperparameters of our models, we performed 5-fold cross-validation using our implementation. The hyperparameter for multinomial naive Bayes is alpha, the smoothing factor. Learning rate, maximum iterations, and the inverse of regularization strength constitute our hyperparameters for Logistic regression. Due to the cross-validation being time-consuming, we used a randomized search as an alternative to an exhaustive search in some cases. The alpha leading to the highest accuracy using multinomial naive Bayes was found to be equal to 0.01 and 0.61 for the 20newsgroup , and IMDb reviews dataset correspondingly. As for logistic regression, a learning rate of 10, a max iteration of 320, and an inverse of regularization strength of 10 achieved the highest accuracy for both datasets. We then evaluated the performance of our models using accuracy and weighted F1 score stated in Table 1. F1 score is an appropriate measure used to balance Precision and Recall when there is an uneven class distribution.

Table 1: Comparison of the two classification models on both datasets

Model	Dataset	Accuracy	F1 Score	5-fold cv accuracy
Multinomial naive Bayes	IMDb review	0.82684	0.82721	0.86564
	20newsgroup	0.70074	0.70643	0.76710
Logistic regression	IMDb reviews	0.8742	0.87420	0.89160
	20newsgroup	0.69078	0.69027	0.75455

As can be seen from Table 1, the logistic regression model outperforms naive Bayes on the IMDb reviews dataset by approximately 5%, resulting in an accuracy of 87%. On the other hand, both models score similarly in performing multiclass classification on the 20newsgroup dataset. It is however noted that the naive Bayes model slightly betters logistic regression by roughly 1%. We compare the performance of models implemented from scratch with the Scikit-learn implementation. It is asserted that our models perform similarly to the Scikit-learn model for the classification on both datasets.

Besides evaluating the performance of our classifiers, we run additional investigations using the ROC curve and confusion matrices. You can find these studies in more detail in the submitted Notebook. The ROC curve of the Logistic regression classifier on the IMDB reviews dataset is illustrated in Figure 3. It can be considered as a performance metric of our classification model and a summary of the key metrics obtained through the confusion matrix in different thresholds. As it can be witnessed in Figure 2, the ROC curve of our logistic regression model is sufficiently close to the ROC curve of a perfect classifier with an AUC of 95%.

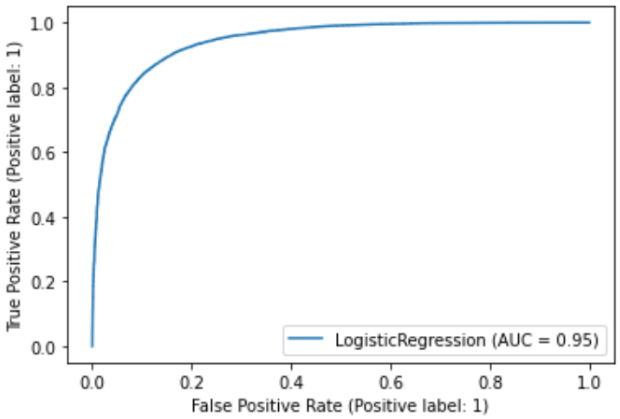


Figure 3: ROC of logistic regression on the IMDb dataset

It is noteworthy that although the number of positive and negative reviews are equal in the IMDb dataset, word counts in positive reviews are larger than negative ones. It can be considered as an uneven dataset that introduces complexities mainly because the input of our model is words and not documents. As can be seen in Figure 4(b), positive reviews have a higher word count compared to negative ones. Interestingly, the number of false positives is higher than the number of false negatives on the IMDb reviews dataset using both models as shown in Figure 5. Besides, the total number of predicted positives (true positives plus false positives) is larger than the total number of predicted negatives (true negatives plus false negatives). The related calculations can be found below for logistic regression. It can be inferred that the trained model tends to predict positive reviews than negative ones.

$$(TP + FP) = 110,000 + 17,000 = 127,000 > (TN + FN) = 110,000 + 15,000 = 125,000$$

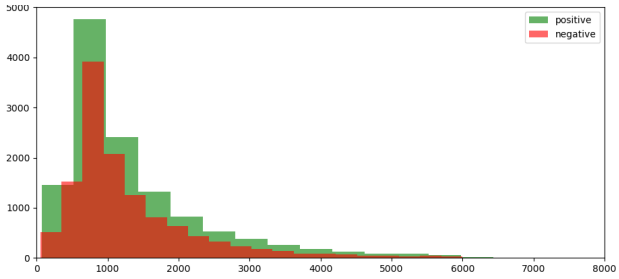
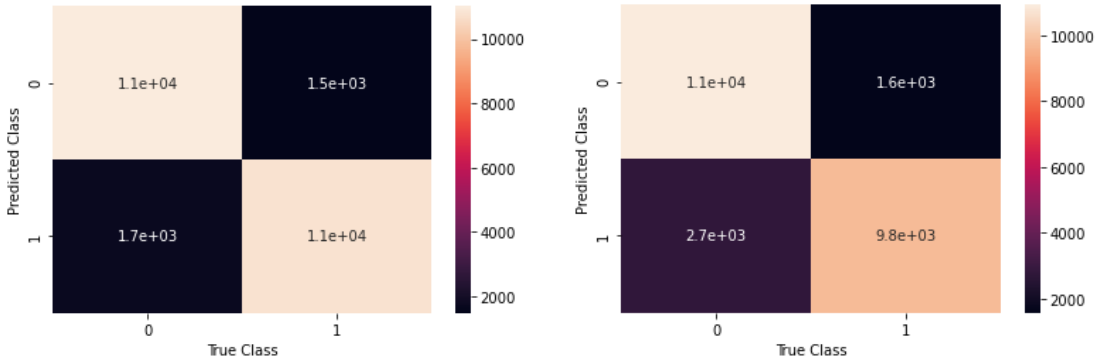


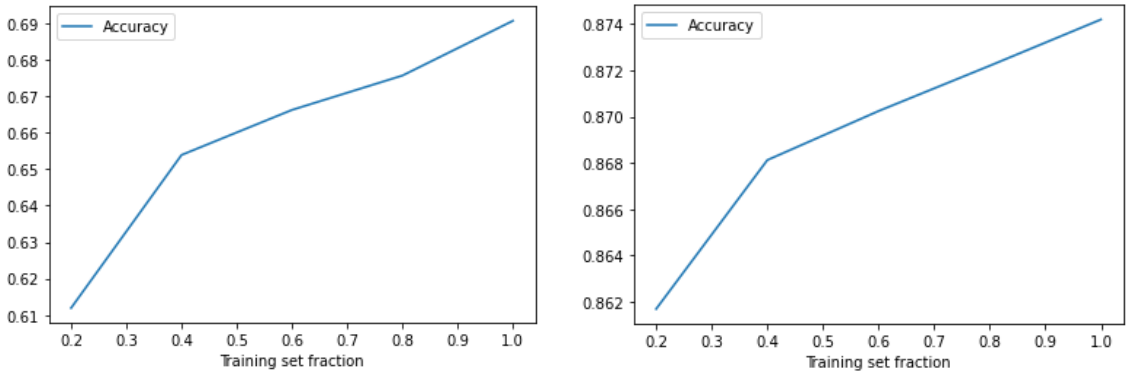
Figure 4: Word count distribution of positive and negative reviews



(a) (b)

Figure 5: Confusion matrix on the IMDb reviews dataset (a) logistic regression (b) multinomial naive bayes

The classification task on both datasets is very time, memory, and energy consuming. To address memory consumption, we implemented our models to directly input and work with sparse matrices. However, gaining higher speed is still a challenge that should be dealt with. To speed up the process of classification, an initial approach would be to reduce the size of the data that the models are trained on. Reducing the training size, would not only lessens features due to having less unique words, but also the number of samples. To analyze this approach, we evaluated the accuracy of our classifiers with 20, 40, 60, 80 and 100% of the training data. The results of this experiment can be seen in Figure 6. Figure 6(a) and 6(b) present the accuracy of logistic regression on the 20newsgroup and IMDb reviews dataset respectively. It can be inferred that while the accuracy of the model on both datasets heightens with the rise of training data, this increase in accuracy is much more significant on the 20newsgroup dataset (8% vs 1.2%). While this is a decent approach to increasing time efficiency on the IMDb dataset, it comes at the detriment of accuracy on the 20newsgroup dataset. Similar results are also observed while performing this experiment on the multinomial naive Bayes classifier which can be found in the Notebook. While reducing

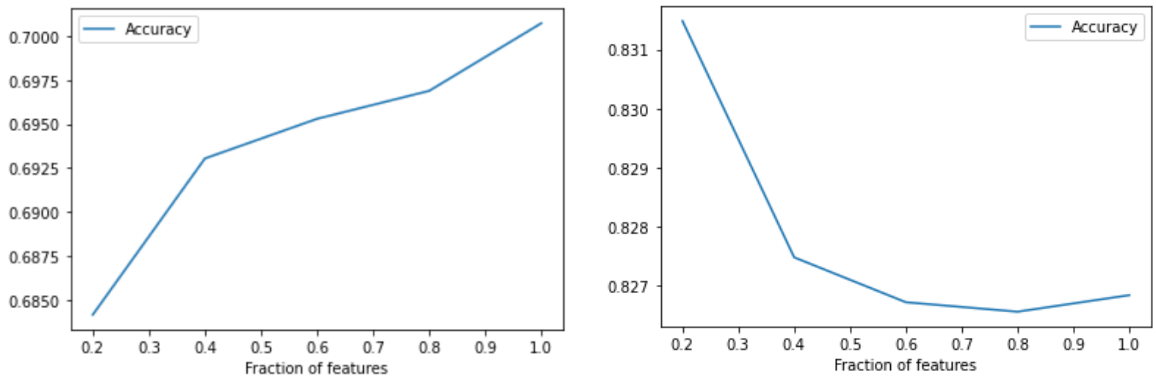


(a) (b)

Figure 6: Training size vs accuracy using logistic regression (a) 20newsgroup dataset and (b) IMDb dataset

the number of training samples diminishes some of the features, many important features become ignored. Therefore, to achieve the same goal of increasing time efficiency while maintaining accuracy high, we reduce the size of the training data by selectively keeping the most significant words (features). Contrary to the previous approach, feature reduction does not

result in important features being ignored. We run this experiment by each time taking 20, 40, 60, 80, and 100% of features, the results of which are shown in Figure 7 for multinomial naive Bayes. We conclude from Figure 7 that in contrast with the previous approach, the accuracy of the models increased with the reduction of features on the IMDb dataset. However, while the accuracy on the 20newsgroup dataset still decreased, this drop in accuracy is not as considerable as the previous experiment (1.2% vs 8%).



(a) (b)

Figure 7: Feature size vs accuracy using multinomial naive Bayes (a) 20newsgroup dataset and (b) IMDb dataset

4 Discussion and Conclusion

To conclude, we preprocessed and vectorized two textual datasets by performing different feature engineering techniques. We then implemented multinomial naive Bayes and logistic regression to conduct multiclass and binary classification on them. After hyperparameter tuning using the implemented k-fold cross-validation, we evaluated the performance of our models. Logistic regression exceeded multinomial naive Bayes on the IMDb reviews dataset while both classifiers performed similarly on the 20newsgroup dataset. Finally, to improve the time efficiency of our models, we examined selective feature and training data reduction. We concluded that selective feature reduction was a better approach to speed up the classifiers. However, a combination of the two mentioned approaches may result in better performance; this can be a subject for future works. Using the "Bag of Words" method for text classification has its downsides. The first and foremost is that the sequence of words is ignored. Alternatives to this method or modifications such as the use of n-grams would be a subject for future investigations.

5 Statement of Contribution

Seyed Yazdan Zinati wrote the code for the models, cross-validation, experiments, visualizations, and preprocessing. Seyedbehnam Mashari wrote the code for the IMDb reader and researched the naive Bayes model with MohammadHadi Foroutan. The report was written by all members.

References

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